

Harry Nguyen

+1 470-667-9000 | nguyenharry2662@gmail.com | linkedin.com/in/harrynguyen26 | github.com/HarryNguyen2662

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • GPA: 3.75/4.0

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Enabled a 3-person startup to run **production-grade AI infrastructure** without a dedicated platform team by owning backend, cloud infrastructure, and ML serving end-to-end as primary engineer
- Restored **100% evaluation success rate** at **sub-50ms p99** by resolving a production **gRPC** deadlock — traced shared resource acquisition conflicts under concurrent calls and redesigned call sequencing across the inter-service layer
- Eliminated **OOM failures** under concurrent production load by architecting a **vLLM/PagedAttention** inference engine with continuous batching, solving KV cache memory fragmentation for LLM serving
- Achieved **99.9% uptime** and cut infrastructure cost **44%** (\$40–60/month) by architecting a multi-AZ ECS Fargate deployment with **Terraform IaC**, CloudWatch observability, and an auto-rollback circuit breaker pattern
- Kept distributed production systems (gRPC, PostgreSQL, ML serving) reliable through on-call ownership — root-caused incidents, then converted findings into runbooks and post-mortem fixes to prevent repeat failures

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Shipped a **C++ IPC transport layer** to Chrome's stable channel — serving **3B+ active users** at **sub-50ms p99** and **10K+ req/sec** — by choosing Protocol Buffers over custom serialization for schema evolution and cross-language compatibility
- Cut settings-navigation p99 latency **96%** (1,200ms down to sub-50ms) with zero production regressions by replacing a mutex-contended search path with a **lock-free concurrent trie**
- Accelerated feature delivery **68%** across **25K+ lines** of Chromium, at **95% test coverage**, by architecting an event-driven TypeScript/React system that decoupled UI state propagation via the observer pattern
- Landed production-ready changes into Chrome's codebase — reviewed and adopted by senior engineers — by delivering design docs and code reviews at **95% test coverage** alongside the Chrome infrastructure team

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Scaled a backend to support **500+ concurrent users** and cut deployment time **90%** by designing a stateless AWS BaaS with JWT auth and a GitHub Actions CI/CD auto-scaling policy
- Cut API response time from 3+ seconds to **sub-100ms** on **10,000+ record** datasets by diagnosing an N+1 query bottleneck and redesigning the data layer with PostgreSQL indexing

PROJECTS

Pulumi — Contributor | Go, TypeScript, Infrastructure-as-Code (IaC)

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge evaluation

Languages: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor